SRI LANKA INSTITUTE OF INFORMATION TECHNOLOGY

Faculty of Computing — Department of Software Engineering

Research Project — Periodic Log Entry Form

Project **R26-SE-009** | Agent 1 | **Nimthara Gunasinha** (IT22078582) | Reporting period **23.03.2026 – 25.08.2026**

1. PROJECT AND STUDENT DETAILS

Project ID	R26-SE-009
Research Topic	Self-Optimized LLM Multi-Agent System for Software Engineering
Supervisor	Prof. Nuwan Kodagoda
Co-Supervisor	Mr. Eishan Weerasinghe
Student Name	Nimthara Gunasinha
Student Registration No.	IT22078582
Individual Component	Agent 1 — Requirements Analyst (AI SRS Engineer) and the SRS user interface
Reporting Period	23 March 2026 to 25 August 2026
Phase Covered	Requirement, implementation and integration phase, representing approximately 65% of the planned project work.

2. SUMMARY OF RP WORK CARRIED OUT DURING THE REPORTING PERIOD

During this reporting period Agent 1 was developed as the requirements and SRS analysis component of the multi-agent system, together with the SRS-facing user interface. The early part of the period established the guided intake flow that turns a plain project idea into structured interview questions, tracked answers, domain inference, stack recommendation, validation and JSON/PDF artefacts. The middle of the period was spent on the SRS front end — intake forms, analysis screens, review tabs and the navigation handoff into the design-selection stage. The later part of the period covered the version 2 rebuild, in which the SRS package was reorganised into a modular application with hardened structured LLM adapters, a multi-source extraction pipeline, domain knowledge catalogues, authenticated plan and SRS endpoints, and a plan-generation runtime prepared for the builder handoff. Group-level contributions were made throughout to the shared AgentForge Studio workspace so that requirements intake, SRS review, artefacts and inter-agent handoff could be demonstrated inside one common interface.

3. DATE-WISE RP WORK (39 ENTRIES, VERIFIED MONTHLY)

MARCH 2026 — 5 ENTRIES

NO.	DATE	RP WORK CARRIED OUT
1	23.03.2026	Defined the scope of Agent 1 as the requirements and SRS analyst within the multi-agent system and identified the outputs needed by later agents.
2	24.03.2026	Reviewed IEEE SRS structure, required requirement sections, and the overall group workflow so Agent 1 could provide a consistent handoff to Agents 2, 3 and 4.
3	26.03.2026	Planned a guided intake flow to transform a plain project idea into structured interview questions, captured answers, and measurable coverage.
4	28.03.2026	Designed section builders for purpose, scope, product functions, user classes, interfaces, and non-functional requirement areas of the SRS.
5	31.03.2026	Added answer normalisation, open-item detection, and completeness tracking concepts to keep the requirement gathering process controlled and reviewable.

MONTHLY VERIFICATION — MARCH 2026

SUPERVISOR'S COMMENTS FOR MARCH 2026	SUPERVISOR'S SIGNATURE	CO-SUPERVISOR'S SIGNATURE
	Signature & Date	Signature & Date

APRIL 2026		9 ENTRIES
NO.	**DATE**	**RP WORK CARRIED OUT**
6	02.04.2026	Added domain inference and preliminary stack recommendation logic so the SRS output could better support downstream development and deployment stages.
7	04.04.2026	Planned support for file-assisted intake using text, PDF, image, and voice-note derived content to enrich requirement capture.
8	07.04.2026	Designed session persistence and artefact delivery so draft and final SRS outputs could be saved, revisited, and exported for later agents.
9	09.04.2026	Defined validation rules for required SRS content, ambiguity reporting, and readiness scoring before the formal handoff to Agent 2.
10	11.04.2026	Coordinated with the group workflow so Agent 1 output could feed shared dashboard, artefact, and project-history views in the common workspace.
11	24.04.2026	Implemented the Agent 1 backend modules in code, including API wiring, session handling, question planning, SRS generation, validation, and JSON/PDF export support.
12	24.04.2026	Integrated Agent 1 with the shared frontend so users can review the generated SRS, functional requirements, interview trace, ambiguity list and stack recommendation.
13	27.04.2026	Refined Agent 1-related shared views and navigation inside AgentForge Studio as part of the group demo consolidation.
14	28.04.2026	Reviewed the end-to-end intake-to-SRS flow with the shared workspace and prepared the reporting-period summary for submission.

MONTHLY VERIFICATION — APRIL 2026		
SUPERVISOR'S COMMENTS FOR APRIL 2026	**SUPERVISOR'S SIGNATURE**	**CO-SUPERVISOR'S SIGNATURE**
	Signature & Date	Signature & Date

MAY 2026		7 ENTRIES
NO.	**DATE**	**RP WORK CARRIED OUT**
15	04.05.2026	Planned the Agent 1 front-end route structure and the screen breakdown for requirement intake, analysis and SRS review inside the shared studio.
16	06.05.2026	Built the Agent 1 SRS screen tab structure and the analysis flow that moves a project from intake through to the generated specification.
17	06.05.2026	Implemented the new-project intake forms with field validation for project brief, domain and constraint capture.
18	08.05.2026	Added the shared Agent 1 components and wired the SRS pages into the common studio navigation.
19	09.05.2026	Completed the front-end styling assets and visual polish for the requirement analysis pages.
20	10.05.2026	Implemented the navigation handoff from Agent 1 into the design-selection stage so approved requirements flow through to Agent 2.
21	11.05.2026	Finalised the Agent 1 front-end build, cleaned the working branch and packaged the reviewed copy into the shared repository.

MONTHLY VERIFICATION — MAY 2026		
SUPERVISOR'S COMMENTS FOR MAY 2026	**SUPERVISOR'S SIGNATURE**	**CO-SUPERVISOR'S SIGNATURE**
	Signature & Date	Signature & Date

JUNE 2026		4 ENTRIES
NO.	**DATE**	**RP WORK CARRIED OUT**
22	09.06.2026	Reviewed the version 1 SRS output against IEEE 830 and ISO/IEC/IEEE 29148 section expectations and recorded the gaps to be closed in the rebuild.
23	16.06.2026	Studied structured-output approaches for LLM-driven specification writing and evaluated how to keep generated sections deterministic and reviewable.
24	21.06.2026	Prepared the interface reference screens for the redesigned SRS workspace and captured the target layouts for the version 2 review.
25	26.06.2026	Planned the module split for the SRS package so intake, interview, planning, generation and document output could evolve independently.

MONTHLY VERIFICATION — JUNE 2026		
SUPERVISOR'S COMMENTS FOR JUNE 2026	**SUPERVISOR'S SIGNATURE**	**CO-SUPERVISOR'S SIGNATURE**
	Signature & Date	Signature & Date

JULY 2026		4 ENTRIES
NO.	**DATE**	**RP WORK CARRIED OUT**
26	07.07.2026	Reworked the SRS service into the shared API and web workspace layout so the component could be served alongside the other three agents.
27	11.07.2026	Carried out the SRS package version change and aligned dependency and interface versions with the integrated application.
28	13.07.2026	Cleaned and consolidated the SRS functions, removing superseded helpers and tightening module boundaries before the rebuild.
29	21.07.2026	Verified the SRS request and response contract shared with the builder side and updated the shared package definitions.

MONTHLY VERIFICATION — JULY 2026		
SUPERVISOR'S COMMENTS FOR JULY 2026	**SUPERVISOR'S SIGNATURE**	**CO-SUPERVISOR'S SIGNATURE**
	Signature & Date	Signature & Date

AUGUST 2026		10 ENTRIES
NO.	**DATE**	**RP WORK CARRIED OUT**
30	20.08.2026	Reviewed the SRS workspace screens — interview, plan review, attachments, diagram viewer and result panes — against the rebuilt backend flow.
31	22.08.2026	Refreshed the SRS package foundation and reorganised it into the modular srs_agent application structure.
32	22.08.2026	Strengthened the project schemas and the repository layer used to persist projects, intake data and generated specifications.
33	22.08.2026	Improved the source extraction pipeline covering PDF, image/OCR, vision and speech-derived requirement input.
34	23.08.2026	Hardened the structured LLM adapters so specification generation returns validated, schema-checked output.
35	23.08.2026	Split the domain knowledge catalogues and expanded the topic and coverage knowledge used for domain classification.
36	23.08.2026	Defined access control and authentication for the plan and SRS endpoints.
37	24.08.2026	Built the plan and SRS assembly stage and refined the clarification and interview flow used to close requirement gaps.
38	24.08.2026	Improved requirement intake and customer-context capture so business background is carried through into the specification.
39	25.08.2026	Split the plan runtime and plan-merge logic and strengthened the plan generation rules ahead of the builder handoff.

MONTHLY VERIFICATION — AUGUST 2026		
SUPERVISOR'S COMMENTS FOR AUGUST 2026	**SUPERVISOR'S SIGNATURE**	**CO-SUPERVISOR'S SIGNATURE**
	Signature & Date	Signature & Date

4. CONSOLIDATED MONTHLY VERIFICATION

MONTH	ENTRIES LOGGED	SUPERVISOR'S SIGNATURE	DATE
March 2026	5		
April 2026	9		
May 2026	7		
June 2026	4		
July 2026	4		
August 2026	10		
Total	**39**		

5. PROJECT WORK PLANNED BEFORE THE NEXT EVALUATION PERIOD

1. Complete the diagram generation runtime and the professional diagram set produced alongside the specification.
2. Finalise the builder handoff contract so approved plans reach Agent 2 with full coverage and traceability data.
3. Raise document generation to the international SRS standard already drafted for the component and validate the output against it.
4. Extend coverage auditing and clarification so requirement gaps are detected and closed before the specification is released.
5. Complete the SRS review, interview and diagram screens in the integrated studio and test them end to end with the other three components.

6. DECLARATION AND SIGNATURES

I certify that the work recorded in this log book for the period 23 March 2026 to 25 August 2026 was carried out by me as the individual component contribution to research project R26-SE-009, and that the group-level contributions recorded here were made jointly with the other project members.

STUDENT Nimthara Gunasinha (IT22078582) — Signature & Date	**SUPERVISOR** Prof. Nuwan Kodagoda — Signature & Date
CO-SUPERVISOR Mr. Eishan Weerasinghe — Signature & Date	**OVERALL SUPERVISOR COMMENTS** Comments